

Stochastic Rupture: Feedback Polarity and Causal Clustering

Code Repository for Manuscript

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Version 1.0 — May 2026

Abstract

This repository contains the complete Python code used to generate all figures in the manuscript entitled “Feedback polarity controls causal clustering and cascade propensity in stochastic rupture”. The code implements a minimal stochastic bond model under displacement control (positive feedback) and force control with transient unloading (negative feedback). It produces analytical scaling curves, synthetic Monte Carlo validation points, network-based avalanche simulations on random graphs, and schematic illustrations of causal synchronization and cross-disciplinary analogies. All figures are generated deterministically (with the exception of synthetically noised demonstration points in Figure 2 and random graph simulations without a fixed seed in Figure 3). The code is provided for reproducibility and further development.

1 Repository structure

The repository contains four Python scripts and four accompanying description files.

Code file	Description file	Figure
Figure_1.py	FIGURE.1.tex	Figure 1: Schematic of three-particle system, tension evolution, and causal synchrony
Figure_2.py	FIGURE.2.tex	Figure 2: Scaling regimes for displacement control, exponential suppression under f
Figure_3.py	FIGURE.3.tex	Figure 3: Minimal coarse-grained spreading simulation on a random Erdős–Rényi g
Figure_4.py	FIGURE.4.tex	Figure 4: Cross-disciplinary analogies: earthquakes, neuronal avalanches, non-recip

2 Requirements

The code requires Python 3.9 or higher and the following Python packages:

```
numpy >= 1.21
matplotlib >= 3.4
scipy >= 1.7
networkx >= 2.6
```

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To install all dependencies, run:

```
pip install numpy matplotlib scipy networkx
```

3 Reproducibility notes

3.1 Figure_1.py (Figure 1)

- **Deterministic:** Yes.
- **Random seed:** Not applicable (no random numbers).
- **Output files:** `fig1_schematic.eps`, `fig1_schematic.png`, `fig1_schematic.pdf`.
- **Notes:** The script uses hard-coded geometric coordinates and analytical tension equations from the manuscript (Eqs. 4, 5, 6). No stochastic elements.

3.2 Figure_2.py (Figure 2)

- **Deterministic:** Partial.
- **Random seed:** `np.random.seed(42)` is set for reproducibility of synthetic Monte Carlo points.
- **Output files:** `fig2_scaling.eps`, `fig2_scaling.png`, `fig2_scaling.pdf`.
- **Notes:** Analytical curves are deterministic. Synthetic Monte Carlo points are generated as $P_c = x/2 + 0.05 \cdot \mathcal{N}(0, 1)$ for visual demonstration only. These do not represent actual simulation output from the stochastic rupture model.

3.3 Figure_3.py (Figure 3)

- **Deterministic:** No.
- **Random seed:** Not set. Graph structure and avalanche outcomes vary across runs.
- **Output files:** None (displays only; does not save to disk).
- **Notes:** For fully reproducible results, users should add `np.random.seed()` and `nx.utils.graph_seed()` before graph generation. The script implements the branching mapping $\mathcal{R} = zp$ where $p = \lambda_{\text{eff}}/2$ for displacement control and $p = r_0 e^{\alpha F} e^{-\alpha F \beta} t_{\text{mat}}$ for force control.

3.4 Figure_4.py (Figure 4)

- **Deterministic:** Yes.
- **Random seed:** Not applicable.
- **Output files:** `fig3_analogies.eps`, `fig3_analogies.png`, `fig3_analogies.pdf`. **Note:** The script saves files with “fig3” in the name; users may wish to rename to “fig4_analogies.*” for consistency with the manuscript.
- **Notes:** Purely schematic; no numerical simulations. All geometric coordinates, colors, labels, and arrow directions are hard-coded.

4 Mathematical mappings

The following key equations from the manuscript are implemented in the code.

4.1 Displacement control (positive feedback)

The causal synchronization probability is:

$$P_c = \left(r_0 e^{-\alpha k \ell_0} + \frac{\alpha k v}{2} \right) t_{\text{mat}} + \mathcal{O}(t_{\text{mat}}^2)$$

In the loading-dominated regime:

$$P_c \sim \frac{\alpha k v}{2} t_{\text{mat}} \equiv \frac{\lambda_{\text{eff}}}{2}$$

4.2 Force control with transient unloading (negative feedback)

The suppression factor is:

$$\frac{P_c}{r_0 e^{\alpha F} t_{\text{mat}}} = e^{-\alpha F \beta}$$

4.3 Branching process mapping

$$\mathcal{R} = z p, \quad p = P_c(t_{\text{mat}} \rightarrow 0)$$

The mean-field threshold for cascading failure is $\mathcal{R} = 1$, corresponding to $\lambda_{\text{eff}} = 2/z$ under displacement control.

5 Output file summary

Script	Output files created
Figure_1.py	fig1_schematic.eps, fig1_schematic.png, fig1_schematic.pdf
Figure_2.py	fig2_scaling.eps, fig2_scaling.png, fig2_scaling.pdf
Figure_3.py	(none – displays only)
Figure_4.py	fig3_analogies.eps, fig3_analogies.png, fig3_analogies.pdf

6 Running the code

To generate all figures, run the scripts in any order:

```
python Figure_1.py
python Figure_2.py
python Figure_3.py
python Figure_4.py
```

Each script will display the figure interactively and save the output files (except Figure_3.py) to the current working directory.

7 License

This code is provided for reproducibility of the research by M. Rajakaruna and H. Rajakaruna. You are free to use, modify, and distribute this code for academic and non-commercial purposes. For commercial use, please contact the corresponding author.

8 Citation

If you use this code or the generated figures in your research, please cite the associated manuscript:

Rajakaruna, M. and Rajakaruna, H. (2026). Feedback polarity controls causal clustering and cascade propensity in stochastic rupture. [Manuscript in preparation/review].

9 Contact

For questions, bug reports, or requests, please contact:

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10 Acknowledgments

This research used computing resources provided by the PraxisModus Institute for Advanced Studies. The manuscript was edited for language clarity using ChatGPT (OpenAI); the authors reviewed all content and take full responsibility for the analysis and conclusions.